The Talking Toy Box

Week 7 Practice - Problem 1 (Basic)

Toy is abstract and assigns a final toyId using one shared static counter in its constructor.
ToyCar and ToyRobot implement different makeSound() messages.

Java solution

```
abstract class Toy {
    private static int nextId = 1001;
    private final String toyId;
    Toy() { toyId = "TOY-" + nextId++; }
    public abstract String makeSound();
    String getToyId() { return toyId; }
}
class ToyCar extends Toy {
    private final String name;
    ToyCar(String name) { this.name = name; }
    public String makeSound() { return name + ": Vroom vroom!"; }
}
class ToyRobot extends Toy {
    private final String name;
    ToyRobot(String name) { this.name = name; }
    public String makeSound() { return name + ": Beep boop!"; }
}
public class Main {
    public static void main(String[] args) {
        ToyCar c = new ToyCar("Speedster");
        ToyRobot r = new ToyRobot("Bolt");
        System.out.println(c.makeSound());
        System.out.println(r.makeSound());
        System.out.println(c.getToyId());
        System.out.println(r.getToyId());
    }
}
```

Sample output

```
Speedster: Vroom vroom!
Bolt: Beep boop!
TOY-1001
TOY-1002
```

Note: Toy itself cannot be instantiated because it is abstract; only concrete toy types can be created.